

Float data type and Read Inputs

Data Type	scanf() Format Specifier	printf() Format Specifier	Example
int	%d	%d	int age = 20;
float	%f	%f / %.2f	float mark = 95.5;
char	%c	%c	char grade = 'A';

Arithmetic Instruction	Result	Arithmetic Instruction	Result
k = 2 / 9	0	a = 2 / 9	0.0
k = 2.0 / 9	0	a = 2.0 / 9	0.2222
k = 2 / 9.0	0	a = 2 / 9.0	0.2222
k = 2.0 / 9.0	0	a = 2.0 / 9.0	0.2222
k = 9 / 2	4	a = 9 / 2	4.0
k = 9.0 / 2	4	a = 9.0 / 2	4.5
k = 9 / 2.0	4	a = 9 / 2.0	4.5
k = 9.0 / 2.0	4	a = 9.0 / 2.0	4.5

Challenge 3: Accepted



Anna University Engineering Cut-off Calculator



A student has scored marks **out of 100** in Mathematics, Physics, and Chemistry.

Write a C program to enter the marks obtained in all three subjects and calculate the engineering admission cut-off.

Formula:

$$\text{Cut-off} = \text{Mathematics} + (\text{Physics} / 2) + (\text{Chemistry} / 2)$$

Display the cut-off mark with **2 decimal places**.

Example Input:

Mathematics = 95

Physics = 90

Chemistry = 85

Example Output:

Engineering Cut-off = 182.50

```
#include <stdio.h>
int main()
{
    float m, p, c, cutoff;
    printf("Enter marks for Mathematics, Physics, Chemistry: ");
    scanf("%f %f %f", &m, &p, &c);
    cutoff = m + (p / 2) + (c / 2);
    printf("Engineering Cut-off = %.2f\n", cutoff);
    return 0;
}
```

GPA Calculator

Write a C program to calculate a student's weighted Grade Point Average (GPA) for three subjects — Mathematics (4 credit hours), Physics (3 credit hours), and English (2 credit hours). The program should take the grade point scored in each subject as input from the user and calculate the overall GPA.

Formula:

$$\text{Weighted GPA} = \Sigma (\text{Grade Point} \times \text{Credit Hours}) / \Sigma (\text{Credit Hours})$$

Sample Input:

Enter grade point for Mathematics: 9

Enter grade point for Physics: 8

Enter grade point for English: 7

Sample Output:

GPA is: 8.11

```
#include <stdio.h>
void main()
{
    int m, p, e;
    int c1 = 4, c2 = 3, c3 = 2;
    float gpa;
    printf("Enter grade points for Mathematics, Physics, English: ");
    scanf("%d %d %d", &m, &p, &e);
    gpa = (float)(m * c1 + p * c2 + e * c3) / (c1 + c2 + c3);
    printf("GPA is: %.2f", gpa);
}
```

Output

Enter grade points for Mathematics, Physics, English: 9 6 7
GPA is: 7.56

Temperature Converter



Write a C program to enter the temperature in **Celsius** and convert it into **Fahrenheit**.

Use the formula:

$$\text{Fahrenheit} = (\text{Celsius} \times 9 / 5) + 32$$

Example Input:

Enter temperature in Celsius: 25

Output:

Temperature in Fahrenheit = 77.00

```
#include <stdio.h>
int main()
{
    double celsius, fahrenheit;

    printf("Enter temperature in Celsius: ");
    scanf("%lf", &celsius);

    fahrenheit = (celsius * 9 / 5) + 32;

    printf("Temperature in Fahrenheit = %.2f", fahrenheit);

    return 0;
}
```

Bus Journey – Seat Calculator



A bus has **12 rows**, with **4 seats in each row**. Passengers occupy the seats **sequentially, starting from the first seat of the first row**.

Write a C program to enter the **number of passengers** and display:

- Total number of seats
- Number of occupied seats
- Number of available seats
- Number of completely empty rows

Example Input:

Enter number of passengers: 32

Output:

Total seats = 48

Occupied seats = 32

Available seats = 16

Completely empty rows = 4

```
#include <stdio.h>

int main()
{
    int rows = 12;
    int seatsPerRow = 4;
    int occupiedSeats;
    int totalSeats, availableSeats, emptyRows;

    printf("Enter occupied seats: ");
    scanf("%d", &occupiedSeats);

    totalSeats = rows * seatsPerRow;
    availableSeats = totalSeats - occupiedSeats;
    emptyRows = availableSeats / seatsPerRow;

    printf("Total seats = %d\n", totalSeats);
    printf("Available seats = %d\n", availableSeats);
    printf("Empty rows = %d", emptyRows);

    return 0;
}
```

Pizza Party – Pizza Calculator



You are organizing a pizza party for your class. Each pizza is cut into **8 equal slices**, and **each student gets one slice**.

Write a C program to **enter the number of students** and calculate:

- **Number of pizzas required**
- **Total number of slices available**

- **Number of leftover slices**

Hint: Since you cannot order part of a pizza, use the following formula to find the number of pizzas:

$$\text{Pizzas Required} = (\text{Number of Students} + 7) / 8$$

Example

Input:

Enter number of students: 18

Output:

Pizzas required = 3

Total slices = 24

Leftover slices = 6

```
#include <stdio.h>
int main()
{
    int students;
    int pizzas, totalSlices, leftoverSlices;

    printf("Enter number of students: ");
    scanf("%d", &students);

    pizzas = (students + 7) / 8;
    totalSlices = pizzas * 8;
    leftoverSlices = totalSlices - students;

    printf("Pizzas required = %d\n", pizzas);
    printf("Total slices = %d\n", totalSlices);
    printf("Leftover slices = %d", leftoverSlices);

    return 0;
}
```